Dattaraj Jadhav

+91 8669366784 email linkedin github

Education

Vishwakarma Institute of Information Technology **Aug 2023 – May 2027**
B.Tech in CSE (AI), CGPA: 8.51 *Pune, Maharashtra*

Relevant Coursework

- Data Structures and Algorithms
- Database Management Systems (DBMS)
- Object-Oriented Programming (OOP)

Summary

B.Tech CSE (AI) student at VIIT Pune with hands-on experience in backend development and Generative AI. Proficient in building REST APIs and real-time systems using Node.js, with strong fundamentals in OOP and DBMS. Experienced in developing RAG pipelines using LangChain, with solid problem-solving skills backed by **200+ LeetCode problems** solved.

Projects

Real-Time Chat Application | *Node.js, Express.js, MongoDB, Socket.IO* **GitHub**

- Developed a real-time chat application with Socket.IO.
- Implemented secure user authentication and authorization including login, logout, and session management using JWT and bcrypt for password hashing.
- Engineered scalable backend APIs with Express.js and MongoDB, supporting dynamic message storage, retrieval, and user session management.
- Designed RESTful API architecture with modular route handling, middleware-based request validation, and error handling to ensure reliable and maintainable server-side logic.

Retrieval-Augmented Document Intelligence System | *TypeScript, LangChain, LangGraph, Supabase, Groq* **GitHub**

- Engineered a two-stage RAG pipeline using **LangGraph** state machines, splitting architecture into optimized *Ingestion* and *Retrieval* graphs with Cross-Encoder re-ranking, reducing query latency on 100-page PDFs from ∼3 min to under 5 seconds.
- Built a fault-tolerant PDF ingestion pipeline with Google GenAI embeddings and **Supabase** (pgvector) semantic search, returning top-5 results with page-level source citations to minimize hallucinations.
- Implemented SSE-based real-time response streaming and a unified LLM abstraction layer supporting multiple providers (Groq, OpenAI, Gemini, Anthropic) via LangChain's model interface.

DocuVerse.AI (Codebase Analyzer) | *Python, FastAPI, Groq LLaMA, MkDocs* **GitHub**

- Built an automated 2-phase LLM documentation pipeline using FastAPI and Groq LLaMA API, generating 3-layer structured MkDocs documentation (module summaries, architecture diagrams, API references) from raw GitHub repositories.
- Implemented AST-based code parsing using Python's native `ast` library to deterministically extract class hierarchies, function signatures, and import graphs — near-eliminating LLM hallucinations on structural metadata by bypassing generative inference entirely.
- Designed concurrent multi-file processing via `ThreadPoolExecutor` with a hash-based `CacheManager` that skips unchanged files on re-runs, reducing prompt context size by ˜89% on average (up to 98% for logic-dense modules) — measured by comparing raw file character counts against AST-extracted prompt payloads across the full codebase.
- Enforced strict Mermaid.js prompt schema (dotted arrow syntax, no inline labels) with an automated post-processing validator (`fix_mermaid_blocks`), significantly improving diagram render reliability and preventing LLM-induced syntax breaks.
- Validated the pipeline across 15+ open-source repositories; architected a `BackgroundTasks`-driven FastAPI backend with in-memory UUID state tracking for safe concurrent job handling and `.zip` payload delivery.

Technical Skills

Languages: Python, JavaScript, TypeScript, C++
Backend: Node.js, Express.js, Prisma ORM, REST APIs
Auth & Security: JWT, bcrypt, Session Management
AI/GenAI: LangChain, Hugging Face, RAG (Retrieval-Augmented Generation), Langgraph
Databases: PostgreSQL, MongoDB, Vector Databases
Real-Time: WebSockets,Socket.IO
Tools & DevOps: Git, GitHub, Postman,Docker
Core CS: Data Structures & Algorithms, OOP, DBMS